Project Title: Hexa-Ludo: An Innovative Hexagonal Board Game

Submitted By:

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1. Project Overview

Project Topic:

Hexa-Ludo is an innovative twist on the traditional Ludo game, redesigned on a hexagonal board with new movement mechanics and power-ups to enhance strategic gameplay. Unlike the conventional square-based board, our version introduces a hexagonal grid where players must navigate efficiently to reach the goal.

Objective:

The primary objective of this project is to develop a strategic AI for the game using heuristic-based decision-making. Additionally, we aim to implement an engaging multiplayer experience with power-ups and new movement rules to increase complexity and strategy.

2. Game Description

Original Game Background:

Ludo is a classic board game played with four players, where each player rolls dice and moves their tokens along a cross-shaped board. The goal is to bring all tokens from the start area to the center of the board while avoiding opponent captures.
Innovations Introduced:

- Hexagonal Board Layout: The game board consists of a 5x5 hexagonal grid, allowing for more diverse movement paths.
- Three-Player Mode: Unlike traditional Ludo, Hexa-Ludo is designed for three players, each controlling a single piece.
- Power-Ups: Certain hexagonal tiles contain power-up spots that grant an extra dice roll upon landing.
- Modified Movement Rules: The movement follows a hexagonal pattern rather than a linear one, increasing strategic depth.

3. AI Approach and Methodology

AI Techniques to be Used:

- Minimax Algorithm: Optimized for multiple players to evaluate the best moves.
- Alpha-Beta Pruning: Enhancing Minimax efficiency by cutting unnecessary calculations.
- Heuristic-Based Movement: Evaluating dice rolls and board positions to make optimal decisions.

Complexity Analysis:

- The game complexity is higher than traditional Ludo due to the hexagonal grid.
- The Minimax AI will have an approximate time complexity of O(b^d), where b is the branching factor and d is the depth of decision-making.

4. Game Rules and Mechanics

Modified Rules:

- Players can move in six directions instead of four.
- Landing on a power-up tile grants an extra turn.
- The game ends when a player reaches position 24.

Winning Conditions:

- The first player to reach the final hexagonal tile (position 24) wins.

Turn Sequence:

1. Player rolls a dice (1-6).
2. The player moves the number of steps shown on the dice.
3. If landing on a power-up tile, the player gets an extra turn.
4. Turn passes to the next player.

5. Implementation Plan

Programming Language: Python

Libraries and Tools:
● Pygame (for GUI and game mechanics)
● Random (for dice rolling and power-up placement)
● NumPy (for AI heuristics and data handling)

Milestones and Timeline:
● Week 1-2: Game design, rule finalization, and board layout development.
● Week 3-4: AI strategy implementation (Minimax, heuristics, and movement logic).
● Week 5-6: Coding and testing game mechanics with GUI.
● Week 7: AI integration, testing, and improvements.
● Week 8: Final game testing and report submission.

6. References

● Pygame Documentation: https://www.pygame.org/docs/
● AI Techniques for Board Games: Research Papers on Minimax and Heuristics
● OpenAI ChatGPT (for conceptual guidance and implementation support)